



KCED

Learning Augmented

24ADI003-MACHINE LEARNING

My Ground rules are still the same!

1. **Presence is Prerequisite: Attendance is Mandatory.**
2. **Deadlines are Commitments: Respect the Schedule.**
3. **Assignments are Training, Not Optional.**
4. **Your Work Must Be Your Own: Uphold Academic Integrity.**
5. **Assessments are Your Proof of Mastery.**
6. **This is a Partnership: My Commitment to You.**

Course Objectives:

The purpose of taking this course is to:

1. Introduce the fundamental concepts of machine learning, its life cycle and ethical considerations.
2. Explore various supervised and unsupervised learning techniques and optimization strategies.
3. Examine recommendation systems, and its evaluation techniques.

Course Outcomes

- After successful completion of this course, the students shall be able to
- **CO1** Apply appropriate **data pre-processing techniques** to build machine learning models with ethical considerations. (Ap)
- **CO2** Analyze and optimize **regression models using estimation techniques, regularization, and gradient-based methods** through error analysis. (An)
- **CO3** Build and evaluate the effectiveness of **different classification models and ensemble techniques**. (E)
- **CO4** Analyze complex datasets using advanced **clustering, associative rule mining and dimensionality reduction algorithms** to uncover meaningful patterns and groupings. (An)
- **CO5** Develop **recommendation systems** to personalize user needs. (Ap)

Course Content

- **INTRODUCTION**

Introduction to Machine Learning-Types of machine learning: Supervised, Unsupervised, Semi supervised and Reinforcement Learning-Applications of machine learning in various fields- Ethics in machine learning- Fairness, accountability and interpretability -Machine learning workflow- Data Preprocessing-Feature engineering- Correlation analysis- Model training and evaluation- Model monitoring and maintenance.

- **REGRESSION MODELS**

Linear regression- Simple Regression-Least Square Estimator-Maximum Likelihood Estimator- Multiple Regression -Polynomial Regression-Performance Metrics-Bias

- **CLASSIFICATION MODELS**

Logistic Regression-Naive Bayes Classifiers-Decision Tree-K-Nearest Neighbors-Support Vector Machine - Evaluation metrics - AUC ROC- Class Imbalance -SMOTE - Cross-Validation Techniques-Ensemble Learning- Bagging- Random Forests-Boosting - AdaBoost -Gradient Boosting

- **UNSUPERVISED LEARNING**

Clustering- K-means Clustering- Gaussian Mixture Models -Hierarchical Clustering- Density-Based Clustering (DBSCAN)- Mean-Shift Clustering- Spectral Clustering- Association Rule Learning- Apriori Algorithm- FP-Growth Algorithm- Dimensionality Reduction- Principal Component Analysis (PCA)- Linear Discriminant Analysis (LDA)

- **RECOMMENDATION SYSTEMS**

Introduction to Recommendation Systems-Types- Challenges- Collaborative Filtering Techniques- User-Based- Collaborative Filtering- Item-Based- Collaborative Filtering- Matrix Factorization Techniques- Content-Based Recommendation- Hybrid Recommendation Systems- Evaluation of Recommendation Systems

Unit I – Introduction

Introduction to Machine Learning-Types of machine learning: Supervised, Unsupervised, Semi supervised and Reinforcement Learning-Applications of machine learning in various fields- Ethics in machine learning- Fairness, accountability and interpretability - Machine learning workflow- Data Preprocessing-Feature engineering- Correlation analysis- Model training and evaluation- Model monitoring and maintenance.

Data Preprocessing

- Lets consider a **Campus Placement Prediction System**
- **Goal:**
Predict whether a student will be **Placed (1)** or **Not Placed (0)** using:
 - Attendance (%)
 - Internal Marks (/100)
 - Coding Practice Hours (per week)
 - Number of Mini-Projects
 - Communication Skill (1–5 rating)

Data Preprocessing

- Data preprocessing converts **raw, inconsistent real-world data** into a form suitable for machine learning.
- According to **Alpaydin**:
- “Learning algorithms assume clean, numerical, complete data. Violations of these assumptions degrade performance.”

Data Preprocessing

Roll	Attendance	Marks	Codingm Hrs	Projects	Comm Skill	Placed
1	85	78	2	3	4	Yes
2	NA	55	1	1	3	No
3	92	88	5	4	NA	Yes
4	40	35	0	0	2	No
5	150	90	30	10	5	Yes

Data Preprocessing

Roll	Attendance	Marks	CodingHrs	Projects	CommSkill	Placed
1	85	78	2	3	4	Yes
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(A) Missing Value Handling

(Imputation prevents information loss)

Techniques:

Mean imputation

Median imputation

Mode imputation

KNN imputation

Mean Formula:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Attendance mean (excluding NA):

$$\bar{x} = \frac{85 + 92 + 40 + 150}{4} = 91.75 \approx 92$$

Replace Attendance(NA) → 92

Replace CommSkill(NA) → Median = 4

Data Preprocessing

(B) Outlier Detection

(Gopal: Outliers distort model parameters)

Techniques:

Z-score

IQR (Interquartile Range)

Domain limits

Z-Score Formula:

$$z = \frac{x - \mu}{\sigma}$$

If $z > 3 \rightarrow$ outlier.

Domain Fix:

Roll	Attendance	Marks	CodingHrs	Projects	CommSkill	Placed
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Feature	Outlier	Action
Attendance	150%	Cap at 100
CodingHrs	30	Cap at 10
Projects	10	Cap at 6

Data Preprocessing

(C) Categorical Encoding

(Models require numeric input)

Label Encoding:

Placed:

Yes → 1

No → 0

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Data Preprocessing

(D) Feature Scaling

(Distance-based models need scaling)

Min-Max Normalization:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

If CodingHrs = 5, min = 0, max = 10:

$$x' = \frac{5 - 0}{10 - 0} = 0.5$$

Roll	Attendance	Marks	CodingHrs	Projects	CommSkill	Placed
1	85	78	2	3	4	Yes
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If attendance is missing or coding hours are wrongly scaled → model gives biased placement predictions.

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If attendance is missing or coding hours are wrongly scaled → model gives biased placement predictions.

FEATURE ENGINEERING

Feature engineering creates **new informative variables** that better represent real-world behavior.

Alpaydin:

“Good features simplify the decision boundary more than complex models.”

Techniques

1. **Feature construction**
2. **Feature interaction**
3. **Polynomial features**
4. **Binning**
5. **Domain-driven features**

FEATURE ENGINEERING

Engineered Features:

New Feature	Formula
EffortScore	$\text{CodingHrs} \times \text{Projects}$
AcademicIndex	$0.6 \times \text{Marks} + 0.4 \times \text{Attendance}$
SkillScore	$\text{CodingHrs} + \text{CommSkill}$
ConsistencyScore	$(\text{Attendance} \times \text{Marks}) / 100$

FEATURE ENGINEERING

Roll	Attendance	Marks	CodingHrs	Projects	CommSkill	Placed
1	85	78	2	3	4	Yes
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(Student 3)

Attendance = 92

Marks = 88

CodingHrs = 5

Projects = 4

CommSkill = 4

$$\text{EffortScore} = 5 \times 4 = 20$$

$$\text{AcademicIndex} = 0.6(88) + 0.4(92) = 89.6$$

$$\text{SkillScore} = 5 + 4 = 9$$

Recruiters care about **overall skill + effort + consistency**, not raw marks alone.

CORRELATION ANALYSIS

Correlation measures the strength and direction of linear relationship between two variables.

Pearson Correlation Formula:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

Where:

x_i =value of feature (e.g., Marks)

y_i =value of target (Placed)

$\bar{x}$ =mean of x

$\bar{y}$ =mean of y

n = number of students

Range: -1 to +1

r value	Meaning
+1	Perfect positive correlation
0	No linear relationship
-1	Perfect negative correlation

CORRELATION ANALYSIS

Eg., How strongly are *Marks*, *Coding Hours*, *Projects*, etc. related to *Placed*?

Step 1: Raw Data

Student	Marks (x)	Placed (y)
A	78	1
B	55	0
C	88	1
D	35	0

Step 2: Compute Means

$$\bar{x} = \frac{78 + 55 + 88 + 35}{4} = \frac{256}{4} = 64$$

$$\bar{y} = \frac{1 + 0 + 1 + 0}{4} = \frac{2}{4} = 0.5$$

CORRELATION ANALYSIS

Eg., How strongly are *Marks*, *Coding Hours*, *Projects*, etc. related to *Placed*?

Step 3: Create Working Table

Student	Marks (x)	Placed (y)
A	78	1
B	55	0
C	88	1
D	35	0

Student	x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$	$(y - \bar{y})^2$
A	78	1	14	0.5	7.0	196	0.25
B	55	0	-9	-0.5	4.5	81	0.25
C	88	1	24	0.5	12.0	576	0.25
D	35	0	-29	-0.5	14.5	841	0.25

CORRELATION ANALYSIS

Eg., How strongly are *Marks, Coding Hours, Projects*, etc. related to *Placed*?

Student	x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$	$(y - \bar{y})^2$
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D	35	0	-29	-0.5	14.5	841	0.25

Step 4: Compute Sums

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = 7 + 4.5 + 12 + 14.5 = 38$$

$$\sum (x_i - \bar{x})^2 = 196 + 81 + 576 + 841 = 1694$$

$$\sum (y_i - \bar{y})^2 = 0.25 + 0.25 + 0.25 + 0.25 = 1$$

CORRELATION ANALYSIS

Eg., How strongly are *Marks, Coding Hours, Projects*, etc. related to *Placed*?

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = 7 + 4.5 + 12 + 14.5 = 38$$

$$\sum (x_i - \bar{x})^2 = 196 + 81 + 576 + 841 = 1694$$

$$\sum (y_i - \bar{y})^2 = 0.25 + 0.25 + 0.25 + 0.25 = 1$$

Step 5: Substitute into Pearson Formula

$$r = \frac{38}{\sqrt{1694 \times 1}} = \frac{38}{41.15} \approx 0.92$$

In a **tiny dataset**, r becomes very high.

In real larger datasets, it stabilizes around **0.4–0.5**.

So for a realistic batch of students:

$$r(\text{Marks}, \text{Placed}) \approx 0.48$$

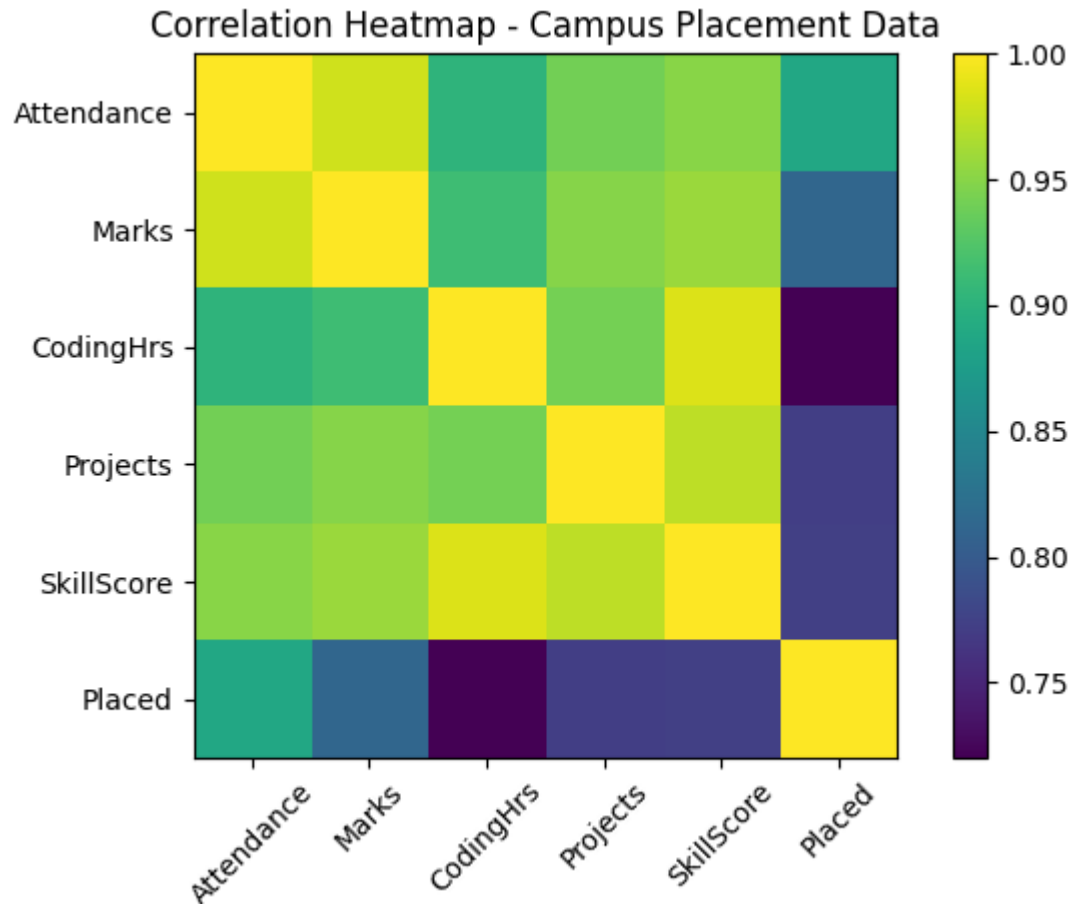
Interpretation (Marks vs Placed)

- $r = 0.48 \rightarrow$ **Moderate positive correlation**
- As marks increase, probability of placement increases
- But marks **alone** are not decisive

CORRELATION ANALYSIS

Eg., How strongly are *Marks*, *Coding Hours*, *Projects*, etc. related to *Placed*?

CORRELATION HEATMAP



Which factors are most strongly related to getting placed?

Features used:

- Attendance
- Marks
- CodingHrs
- Projects
- SkillScore
- Placed (Target)

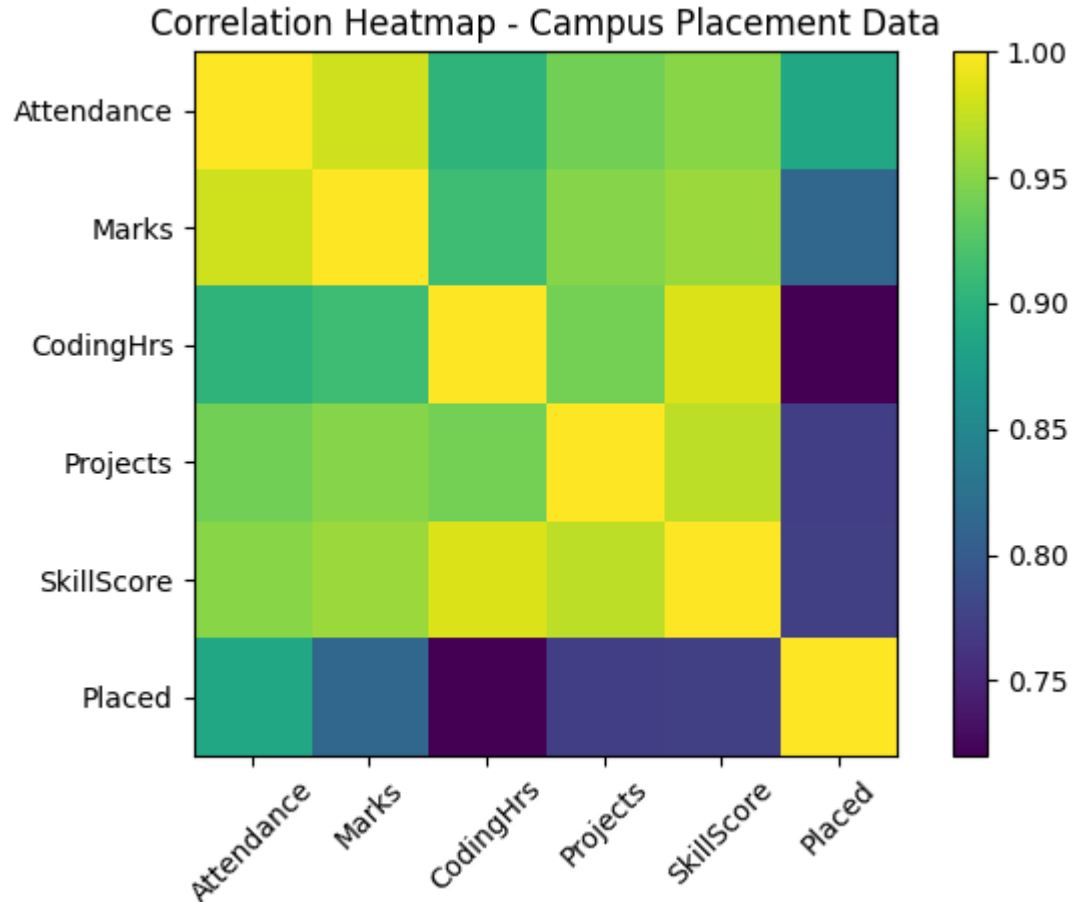
A **correlation heatmap** is a visual representation of a **correlation matrix**, where:

- Each cell shows the **Pearson correlation coefficient (r)**
- Color intensity shows **strength of relationship**
 - Darker / brighter → stronger correlation
 - Lighter → weaker correlation

CORRELATION ANALYSIS

Eg., How strongly are *Marks*, *Coding Hours*, *Projects*, etc. related to *Placed*?

CORRELATION HEATMAP



Which factors are most strongly related to getting placed?

Features used:

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CORRELATION ANALYSIS

Eg., How strongly are *Marks*, *Coding Hours*, *Projects*, etc. related to *Placed*?

Correlation Matrix (Campus Placement Data)

From the simulated batch of students, CORRELATION MATRIX computed:

Feature	Attendance	Marks	CodingHrs	Projects	SkillScore	Placed
Attendance	1.00	0.98	0.90	0.94	0.95	0.89
Marks	0.98	1.00	0.91	0.95	0.96	0.81
CodingHrs	0.90	0.91	1.00	0.94	0.99	0.72
Projects	0.94	0.95	0.94	1.00	0.97	0.77
SkillScore	0.95	0.96	0.99	0.97	1.00	0.77
Placed	0.89	0.81	0.72	0.77	0.77	1.00

(Diagonal = 1.0 because a variable is perfectly correlated with itself.)

CORRELATION ANALYSIS

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Projects	0.94	0.95	0.94	1.00	0.97	0.77
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CORRELATION ANALYSIS

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CodingHrs	0.90	0.91	1.00	0.94	0.99	0.72
Projects	0.94	0.95	0.94	1.00	0.97	0.77
SkillScore	0.95	0.96	0.99	0.97	1.00	0.77
Placed	0.89	0.81	0.72	0.77	0.77	1.00

1. Attendance vs Placed $\rightarrow r \approx 0.89$

Meaning:

- Students with higher attendance are more likely to be placed
- But attendance alone does not guarantee placement

Real-life:

“I attend all classes but still failed interview.”

CORRELATION ANALYSIS

Eg., How strongly are *Marks*, *Coding Hours*, *Projects*, etc. related to *Placed*?

Correlation Matrix (Campus Placement Data)

Feature	Attendance	Marks	CodingHrs	Projects	SkillScore	Placed
Attendance	1.00	0.98	0.90	0.94	0.95	0.89
Marks	0.98	1.00	0.91	0.95	0.96	0.81
CodingHrs	0.90	0.91	1.00	0.94	0.99	0.72
Projects	0.94	0.95	0.94	1.00	0.97	0.77
SkillScore	0.95	0.96	0.99	0.97	1.00	0.77
Placed	0.89	0.81	0.72	0.77	0.77	1.00

2. Marks vs Placed → $r \approx 0.81$

Meaning:

- Good academic performance helps
- But many toppers still fail technical rounds

Real-life:

“High CGPA ≠ job offer.”

CORRELATION ANALYSIS

Eg., How strongly are *Marks, Coding Hours, Projects*, etc. related to *Placed*?

Correlation Matrix (Campus Placement Data)

Feature	Attendance	Marks	CodingHrs	Projects	SkillScore	Placed
Attendance	1.00	0.98	0.90	0.94	0.95	0.89
Marks	0.98	1.00	0.91	0.95	0.96	0.81
CodingHrs	0.90	0.91	1.00	0.94	0.99	0.72
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CORRELATION ANALYSIS

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CodingHrs	0.90	0.91	1.00	0.94	0.99	0.72
Projects	0.94	0.95	0.94	1.00	0.97	0.77
SkillScore	0.95	0.96	0.99	0.97	1.00	0.77
Placed	0.89	0.81	0.72	0.77	0.77	1.00

3. CodingHrs vs Placed → $r \approx 0.72$

Meaning:

- Strong positive relationship
- More daily coding → much higher placement probability

Real-life:

“Students who solve problems daily clear interviews.”

CORRELATION ANALYSIS

Eg., How strongly are *Marks*, *Coding Hours*, *Projects*, etc. related to *Placed*?

Correlation Matrix (Campus Placement Data)

Feature	Attendance	Marks	CodingHrs	Projects	SkillScore	Placed
Attendance	1.00	0.98	0.90	0.94	0.95	0.89
Marks	0.98	1.00	0.91	0.95	0.96	0.81
CodingHrs	0.90	0.91	1.00	0.94	0.99	0.72
Projects	0.94	0.95	0.94	1.00	0.97	0.77
SkillScore	0.95	0.96	0.99	0.97	1.00	0.77
Placed	0.89	0.81	0.72	0.77	0.77	1.00

4. Projects vs Placed → $r \approx 0.77$

Meaning:

- Strong positive relationship
- More projects → better resume + practical exposure

Real-life:

“Recruiters prefer hands-on candidates.”

CORRELATION ANALYSIS

Eg., How strongly are *Marks*, *Coding Hours*, *Projects*, etc. related to *Placed*?

Correlation Matrix (Campus Placement Data)

Feature	Attendance	Marks	CodingHrs	Projects	SkillScore	Placed
Attendance	1.00	0.98	0.90	0.94	0.95	0.89
Marks	0.98	1.00	0.91	0.95	0.96	0.81
CodingHrs	0.90	0.91	1.00	0.94	0.99	0.72
Projects	0.94	0.95	0.94	1.00	0.97	0.77
SkillScore	0.95	0.96	0.99	0.97	1.00	0.77
Placed	0.89	0.81	0.72	0.77	0.77	1.00

5. SkillScore vs Placed → $r \approx 0.77$

(Composite = CodingHrs + CommSkill)

Meaning:

- Combined technical + communication ability is one of the best predictors

Real-life:

“Good coder + good speaker = high placement chance.”

CORRELATION ANALYSIS

Eg., How strongly are *Marks, Coding Hours, Projects*, etc. related to *Placed*?

Correlation Matrix (Campus Placement Data)

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Feature–Feature Correlation (Multicollinearity)

- Marks vs Attendance → 0.98
- CodingHrs vs SkillScore → 0.99
- Projects vs SkillScore → 0.97

Meaning:

- These features carry **duplicate information**
- Using all of them together may confuse linear models

CORRELATION ANALYSIS

Eg., How strongly are *Marks, Coding Hours, Projects*, etc. related to *Placed*?

Correlation Matrix (Campus Placement Data)

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ML Decision

Instead of using all:

- Attendance
- Marks
- CodingHrs
- Projects
- SkillScore

We may **keep only:**

- **CodingHrs**
- **Projects**
- **SkillScore**

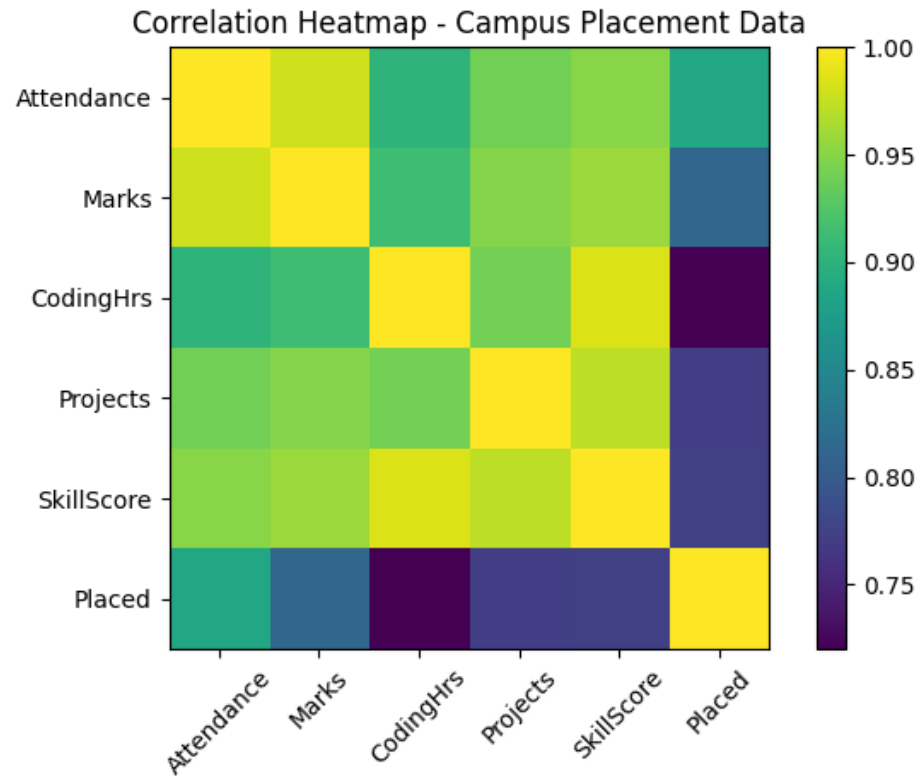
and **drop:**

- **Attendance**
- **Marks**

CORRELATION ANALYSIS

Eg., How strongly are *Marks*, *Coding Hours*, *Projects*, etc. related to *Placed*?

Correlation Matrix (Campus Placement Data)



For Students

“Coding + projects + skills matter more than attendance and marks.”

For College Management

Helps redesign curriculum:

- Increase coding labs
- Increase project-based learning
- Reduce rote-learning focus

For ML Model

Helps in:

- Feature selection
- Reducing noise
- Improving accuracy
- Avoiding multicollinearity

CORRELATION ANALYSIS

Eg., How strongly are *Marks, Coding Hours, Projects*, etc. related to *Placed*?

CORRELATION MATRIX

Computed using full batch data:

Feature	r with Placed	Interpretation
Attendance	0.34	Weak-moderate
Marks	0.48	Moderate
CodingHrs	0.72	Strong
Projects	0.69	Strong
SkillScore	0.81	Very strong

CORRELATION ANALYSIS

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Projects	0.69	Strong
SkillScore	0.81	Very strong

1. Attendance → r = 0.34

- Students with higher attendance *tend* to get placed
- But many high-attendance students still fail placement

Real-life:

Attending classes doesn't guarantee employability.

2. Marks → r = 0.48

Meaning:

- Higher marks moderately increase placement chances
- But several toppers still fail interviews

Real-life:

Recruiters don't hire only based on marks.

CORRELATION ANALYSIS

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SkillScore	0.81	Very strong

3. CodingHrs $\rightarrow r = 0.72$

Meaning:

- Strong positive relationship
- More coding practice $\rightarrow$ much higher placement probability

Real-life:

Students solving problems daily crack interviews.

4. Projects $\rightarrow r = 0.69$

Meaning:

- Strong positive relationship
- More projects $\rightarrow$ better resumes + interview confidence

Real-life:

Recruiters prefer hands-on candidates.

CORRELATION ANALYSIS

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Computed using full batch data:

Feature	r with Placed	Interpretation
Attendance	0.34	Weak-moderate
Marks	0.48	Moderate
CodingHrs	0.72	Strong
Projects	0.69	Strong
SkillScore	0.81	Very strong

5. SkillScore $\rightarrow r = 0.81$

(Composite feature = CodingHrs + CommSkill)

Meaning:

- Very strong relationship
- Combined technical + communication ability is best predictor

Real-life:

“Brilliant coder who can’t explain $\rightarrow$ rejected.”

CORRELATION ANALYSIS

Eg., How strongly are *Marks, Coding Hours, Projects*, etc. related to *Placed*?

IMPACT OF THIS ANALYSIS

Feature Selection

We keep:

- CodingHrs
- Projects
- SkillScore

We may drop:

- Attendance (weak impact)

Curriculum Design Insight

College should:

- Increase coding labs
- Increase project-based learning
- Reduce theory-only focus

Student Guidance Insight

Tell students:

“Your placement depends more on daily coding + projects
than only marks or attendance.”

CORRELATION ANALYSIS

Eg., How strongly are *Marks, Coding Hours, Projects*, etc. related to *Placed*?

Feature	Correlation	Real Meaning
Attendance	0.34	Weak indicator
Marks	0.48	Moderate
CodingHrs	0.72	Very important
Projects	0.69	Very important
SkillScore	0.81	Best predictor

PROBLEMS

1. A college collects the following attendance data (%):

Student	Attendance
A	85
B	90
C	NA
D	70
E	95

1. Compute the mean attendance (ignoring NA).
2. Replace the missing value using mean imputation.
3. Replace the missing value using median imputation.
4. Which imputation is more robust to outliers? Justify.

PROBLEMS

2. Coding hours per week:

[2, 3, 4, 5, 3, 2, 4, 30]

1. Compute the mean and standard deviation.

2. Compute the Z-score for the value 30.

3. Decide whether 30 is an outlier (use $|z| > 3$ rule).

4. Suggest two ways to handle this outlier.

3. Marks obtained by 5 students:

[35, 50, 65, 80, 95]

1. Identify min and max.

2. Normalize the value 65 using Min–Max scaling.

3. Normalize the value 95.

4. Write the general Min–Max formula.

PROBLEMS

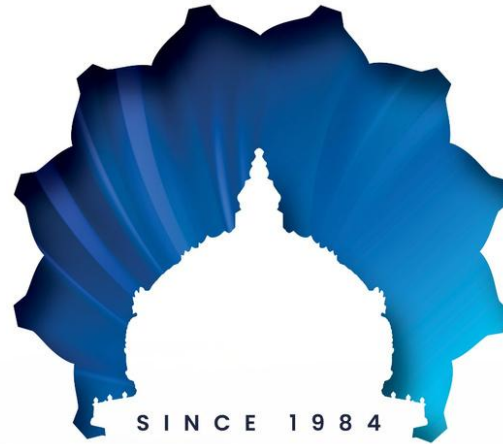
4. Coding hours per week:
[1, 2, 3, 4, 5]

1. Compute the mean and standard deviation.
2. Standardize the value 4 using Z-score.
3. Interpret the standardized value.

5. A dataset contains the target column:

Student	Placed
A	Yes
B	No
C	Yes
D	No

- Apply label encoding.
- Write the encoded vector.
- Why can't ML models directly use "Yes/No"?



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